- Take a spot of red paint, add a little bit of green paint to it. The change in colour cannot be seen immediately. Add a little more, now, you can see a slight change in the colour. A little more ….., Hey! don't you see a different colour now? You add green paint drop by drop to red paint, you finally get a new colour. The activities explained above follow ***iterative processes***.



Hence, an ***iterative process*** is a procedure that is repeated many times which gives rise to a new form.

MATHEMATICS ALIVE – INFORMATION PROCESSING IN REAL LIFE



Fibonacci sequence in nature



Orderly arrangement of fruits in the shop

5.2 Iterative Process in Numbers

The above iterative processes can be seen in our daily life. It can be seen in number sequences also. The numbers may increase or decrease following a pattern.

1. Observe the following sequences and find the pattern that generates each one of them.
   - 1, 3, 5, 7, ... The pattern which generates these numbers is 1, 1+2, 3+2, 5+2...
   - 50, 48, 46, 44,... The pattern which generates these numbers is 50, 50-2, 48-2, 46-2...
   - 2, 4, 6,... The pattern which generates these numbers is 1x2, 2x2 ; 3x2, ...
   - 1, 4, 9, 16... The pattern which generates these numbers is 1x1, 2x2 , 3x3...
   - 2, 6, 12, 20, 30,... The pattern which generates these numbers is 1x2, 2x3, 3x4, 4x5, ...
   - 2, 4, 8,16,... The pattern which generates these numbers is 2x1, 2x2, 2x2x2...
2. Observe the pattern, 1, 10, 100, … . When the number of zeros increase the value also increases.

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